



Made in PESU for PESU ❤️ · Capstone 2026-28

Capstone team formation, done right.

A web portal that replaces the WhatsApp-group chaos of finding a project team with a system where the university's rules enforce themselves — and finding the *right* teammates takes seconds, not weeks.

1,500+

STUDENTS SUPPORTED

<40ms

PER REQUEST

58

FACULTY MENTORS

1 file

WHOLE DATABASE

Why TeamUp



Today: WhatsApp chaos

Every student needs a 4-person team under strict rules (grade mix, branch, gender). Finding teammates over group chats is messy, and you can't tell if a team is even *valid* until it's too late.



TeamUp: rules that enforce themselves

An invalid team is *impossible to form*. Live filters, a searchable directory, faculty mentors, and an AI assistant — all on one page, hosted on the college server.

FEATURE SET

Everything a batch needs



Rule-enforced teams

Exactly 4 members; grade mix must be AABC/ABBC/ABCC/AACC/BBCB; CSE+AI/ML combine; ECE-only ECE teams. Checked at every step.



Browse & filter

Teams by branch, domain, open grade slot, "eligible to join". Live "6 of 9 teams" count.



Student directory

Search 1,500+ students by name/SRN; see who's free; filter by branch, gender, grade, interest.



Requests & invites

Request to join, or invite eligible students. Any member can act; WhatsApp shared to vibe-check.



Leadership tools

Approve, kick, disband, transfer the crown, team description — clear role permissions.



Faculty mentors

Real PES faculty with photos, emails, domain expertise. Searchable; attach to your team.



AI Assistant

"Which team is Priya in?" · "Mentors for Blockchain?" — answers from live data, never hallucinated.



Privacy & security

CGPA never shown (only A/B/C); script passwords; email-OTP signup & reset; activity log.



Mobile-ready

Fully responsive — built for the phones most students will actually use.

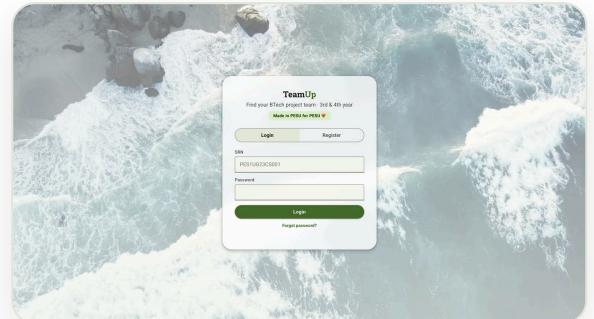
Getting in & finding a team

1

Secure sign-in

Students log in with their SRN. Registration is confirmed by an **email OTP**, and forgotten passwords reset the same way.

- ▶ Salted script password hashing
- ▶ Email-verified accounts, unique per email
- ▶ Multi-device sessions



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One profile, everything captured

Name, SRN, branch, gender and CGPA at signup — plus an email for recovery and an optional WhatsApp number so teammates can reach out.

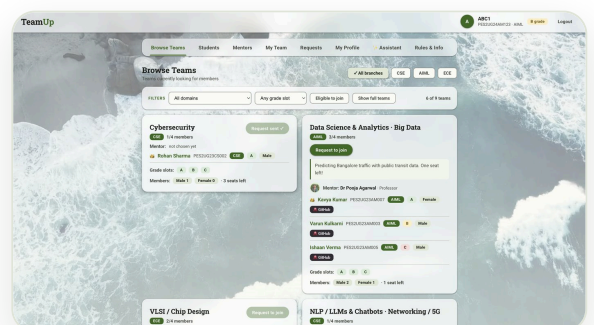
- ▶ CGPA becomes a private A/B/C grade to others
- ▶ Branch drives who you can team with

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Browse teams looking for members

Every open team on one screen, filtered by branch, domain and open grade slot. Cards show live seat counts and exactly which grades still fit.

- ▶ "Eligible to join" shows only teams you can join
- ▶ Grade-slot logic enforces the allowed combos



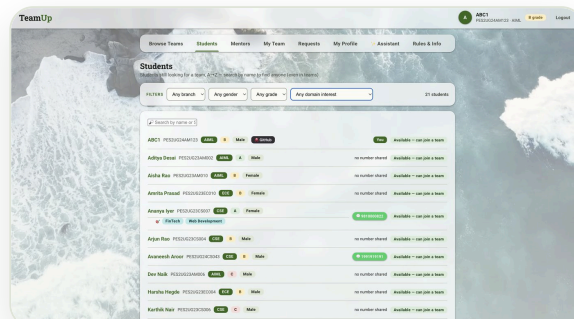
Finding people & forming the team

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Search the whole batch

An A→Z directory of every student — see who's still free, filter by branch, gender, grade or domain interest, and request to join their team on the spot.

- ▶ Runs in SQL — instant even at 1,500 students
- ▶ Interests & contact shown inline

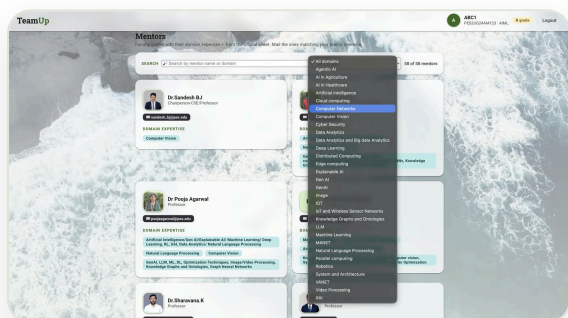


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Pick the right mentor

All 58 EC-campus faculty with photos, emails and domain expertise — searchable by name or domain, so you find a guide whose interests match your project.

- ▶ Parsed from the official faculty-domains sheet
- ▶ One click to attach a mentor to your team

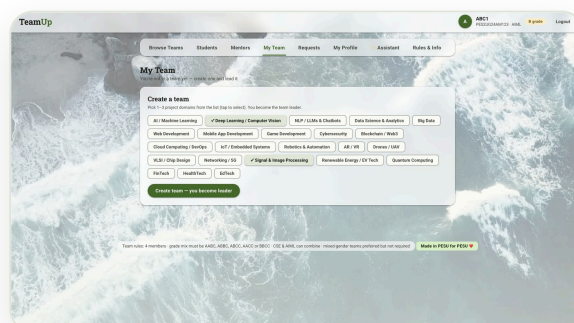


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Run your team

The leader adds a description and mentor; members are listed with grades and GitHub. Leadership can be transferred, members removed, and open slots shown live.

- ▶ Any member can accept requests & invite students
- ▶ Kick / disband / transfer are leader-only

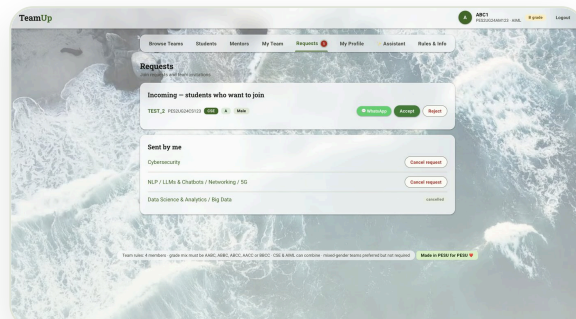


Requests, profile & the AI assistant

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Requests & invitations

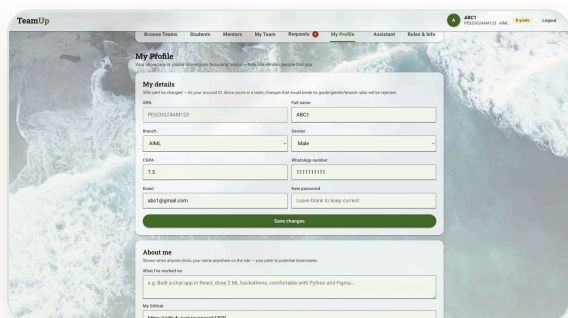
Incoming join requests with a WhatsApp button to chat first, Accept / Reject, and a running list of the requests you've sent — cancel any of them anytime.



8

Your showcase profile

Edit your details, write an "about me", add your GitHub, pick interested domains, and list past projects — everything teammates see when they open your name.

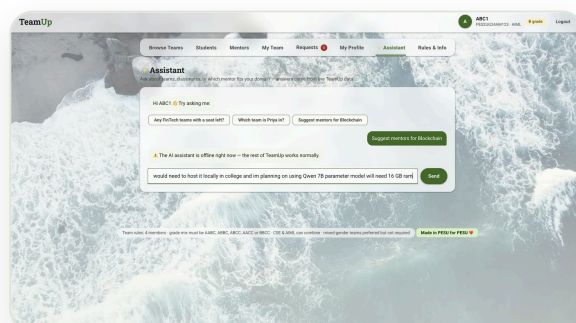


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Ask the AI assistant

A chatbot that answers "which teams work on FinTech?", "which team is Priya in?", or "suggest mentors for Blockchain" — using **tool-calling over the live database**, so answers are always real.

- Runs on a small local model — no GPU, no data leaves campus



BUILT TO LAUNCH

Engineered for the whole batch

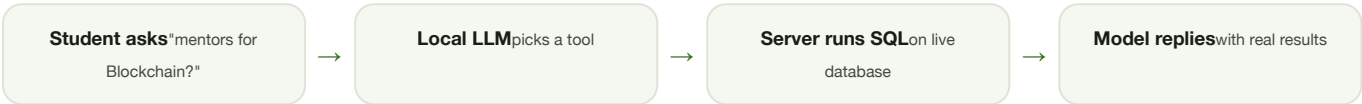
**Load-tested**
1,600 students / 350 teams — 20–40 ms/request; 200 concurrent requests served in ~3 s.

**Crash-safe**
SQLite + WAL, ACID transactions. Verified with `kill -9` mid-write — zero data loss.

**Backed up**
Nightly SQLite snapshots, 14-day retention. One file to copy, restore, or hand off.

HOW THE AI ASSISTANT WORKS

Answers are live data, never invented



WHY SQLITE, NOT MONGODB

No database server to host

- ✓ It's a **file**, not a service — nothing for IT to install, expose a port for, or maintain
- ✓ Real relational SQL — tables, joins, ACID transactions
- ✓ Portable further — exports to MySQL / PostgreSQL with zero rework

TECH STACK

- ✓ Node.js + Express — one service, no build step
- ✓ SQLite (better-sqlite3) — one indexed file
- ✓ Vanilla HTML / CSS / JS frontend
- ✓ Ollama + Qwen for the assistant (CPU, no GPU)
- ✓ Docker image · systemd · Caddy HTTPS

SERVER REQUIREMENTS

	Without AI	With AI
CPU	4 cores	4–8 cores
RAM	8 GB	16 GB
Disk	25 GB	30 GB
GPU	none	none needed

HOSTING = 3 COMMANDS

```
docker run -d -p 3000:3000 \
-v /opt/teamup/data:/data \
avaneesharoor/teamup

curl -fsSL ollama.com/install.sh | sh
ollama pull qwen3:4b
```

One process to host, one file for the database — the entire operational footprint.